

Parth Bhagwat

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EDUCATION

University of California, San Diego

Expected June 2027

Bachelor of Engineering in Mechanical Engineering

GPA 3.57

- **Relevant Coursework:** Signals and Systems, Differential Equations, Vector Calculus, Mechanical Design, Linear Controls, Solid Mechanics, Computational Methods, Robotics, Materials Science, Fluid Mechanics, Thermodynamics

Leadership: UC San Diego Club Cricket President, Sigma Alpha Mu Fraternity – Gamma Rho Chapter President

PROFESSIONAL EXPERIENCE

Indian Space Research Organization (ISRO) – Remote Sensing

New Delhi, India

Intern

Jul 2025 – Aug 2025

- Co-Authored on a research paper titled “Novel GeoAI Framework for Conversational Satellite Analysis using LangGraph and Google Earth Engine.”
- Simplified satellite image retrieval via LangGraph to condense multi-step Google Earth Engine scripting into a single-line query in order to simplify and streamline the process of accessing satellite imagery for further analysis.
- Achieved 98% accuracy in RGB and simple change detection image generation across 50 geographic locations to validate the robustness of the query-to-image pipeline, whilst also allowing AI to interpret more complex queries accurately.
- Enabled non-technical users to access global satellite imagery without writing code enabling usability across departments.
- Reduced manual coding time by 90% and eliminated coordinate lookup to streamline workflows for scientists in research.

Nano Bioelectronic (NBE) Lab

San Diego, CA

Research Assistant

Feb 2026 – Present

- Conducted finite-element static stress element simulation in Autodesk Fusion to evaluate the behavior of a mouthguard-based bioelectronic device under mechanical loads ensuring the design could withstand realistic usage conditions.
- Developed and executed COMSOL Multiphysics CFD simulations to model liquid flow through porous materials allowing the lab to understand expected velocity and concentration to establish theoretical baselines for comparisons.
- Generated simulation-driven insights across mechanical and fluid domains to support UCSD NBE Lab’s bioelectronic wearable research, helping the team to identify discrepancies between theory and experiment.

Rostrum Grand Limited

Hong Kong

Intern

Jul 2024 – Sep 2024

- Analyzed and sorted large datasets of financial returns, ensuring accuracy and consistency across various reports and databases, allowing the company to gather data to further process.
- Consolidated data into a centralized system to enable the company to generate graphs and perform statistical analyses for enhanced data-driven insights, which would allow clients to make informed decisions based on a large set of data.

PROJECTS

Robotic Arm

San Diego, CA

Apr 2025 – Jun 2025

- Designed a 360-degree multi-DOF robotic arm in Autodesk Fusion for ring pick-and-place optimizing kinematics and precise operation achieving 85% success rate over 20 trials.
- Conducted risk assessments and performance calculations using force and power analysis and static simulations on Fusion to ensure the arm’s structural integrity and operational reliability under expected loading conditions.
- Fabricated and assembled the robotic arm using laser cutting, 3D printing, and shop tools to produce a robust competition-ready system capable of consistently executing its intended tasks.

TritonUAS

San Diego, CA

Sep 2024 – Jun 2025

- Collaborated with a team to design and build an autonomous aircraft capable of delivering payloads to precise locations without human intervention for a competition against other universities across the US.
- Developed prototypes for the control surface using Onshape, laser cutting, and 3D printing, ensuring that the optimal design was used for the final design, whilst competition standards were met and structural integrity was maintained.

SKILLS

Robotics, Deep Learning, Sim-to-Real, Python, MATLAB, Linux, Onshape, Autodesk Fusion, COMSOL Multiphysics, LangChain, LangGraph, SolidWorks, Google Earth Engine, Competitive Market Research and Analysis, Growth Strategies, GPS, ROS2, Embedded Systems
